

Project #1 Proposal

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For project #1, I would like to focus on low earth orbit refueling for interplanetary spacecrafts. With the increase in large spacecrafts attempting to cross earth orbits, fuel depots are bound to be more ubiquitous. Naturally, it makes sense to understand the dynamics of docking, transferring propellant and releasing from a fuel depot in low earth orbit. Specifically for this project, two spacecraft of different geometries will be simulated along with drag, SRP, and J2. The altitude of the spacecraft will be varied from 500 km to 1000 km to understand the effects of these perturbations on the relative motion of the depot and the spacecraft. Additionally, the relative offset will be varied.

Here's a list of tasks:

- Define interplanetary spacecraft and fuel depot characteristics:
 - Shape, size, features (solar arrays, etc.)
 - Orbital parameters
 - Orbit semi-major axis, eccentricity, inclination, RAAN, AOP, true/mean anomaly
 - Initial spacecraft & fuel depot offset
 - This could be a simple true anomaly/semi-major axis offset
 - Fuel transfer duration
- Create a simple simulation with two spacecrafts with unperturbed motions.
 - To simplify the simulation, assume that the sun is stationary in the +X inertial reference frame.
- Integrate drag, SRP, J2 perturbations into the simulation.
- Vary the following parameters and note observations
 - Altitude
 - Initial offset (could be true anomaly or semi-major axis)
 - Propellant transfer time

The drag and SRP models will be used from Space Mission Analysis and Design.

Here's a list of references that could be useful for this project:

- [Risk Analysis of On-Orbit Spacecraft Refueling Concepts](#)
- [On Orbit Satellite Servicing Study](#)
- [On-Orbit Servicing: Inspection, Repair, Refuel, Upgrade, and Assembly of Satellites in Space](#)